

Zimo Wang

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EDUCATION

- University of California, San Diego** San Diego, U.S.A.
 - *Data Science Ph.D. Candidate, Advisor: Tzu-Mao Li* Sep 2023 - Present
 - M.S. in Data Science (Master's along the way)* 2023–2026
- Zhejiang University** Hangzhou, P.R.C.
 - *Bachelor of Science, Geographical Information Science (**Rank 1st**);* Sep 2019 - June 2023
 - Overall GPA: 3.97/4.00, 90.24/100*
- Zhejiang University** Hangzhou, P.R.C.
 - *Double Major, Statistics (**Rank 1st**); Minor, Computer Sci. and Tech.* Sep 2019 - June 2023

HONORS AND AWARDS

- National Scholarship, Ministry of Education of P.R. China 2020
- Zhejiang University Scholarship - First Prize, Third Prize, and Second Prize 2020-2022
- Academic Excellence Award, Zhejiang University 2020-2022
- Outstanding Student, Zhejiang University 2020

PUBLICATIONS AND PREPRINTS

- **Zimo Wang**, Junkun Yuan, Angtian Wang, Haotian Yang, Canyu Zhang, Siyuan Yuan, Xingchang Huang, Bo Liu, Yizhi Wang, Yiding Yang, Chongyang Ma, Gordon Guocheng Qian. **PDMD: Projected Distribution Matching Distillation for Video Diffusion Models**. *arXiv preprint*. 2026.
- **Zimo Wang**, Ishit Mehta, Haolin Lu, Xunpeng Huang, Chung-En Sun, Ge Yan, Tsui-Wei Weng, Tzu-Mao Li. **Distance Marching for Generative Modeling**. *Conference on Neural Information Processing Systems*. 2026.
- **Zimo Wang***, Cheng Wang*, Taiki Yoshino, Sirui Tao, Ziyang Fu, Tzu-Mao Li. **HotSpot: Signed Distance Function Optimization with an Asymptotically Sufficient Condition**. *Conference on Computer Vision and Pattern Recognition (**Highlight**)*. Published. 2025. (* Equal contribution.)
- Chaoran Tian, Weihong Pan, **Zimo Wang**, Mao Mao, Guofeng Zhang, Hujun Bao, Ping Tan, Zhaopeng Cui. **DPS-Net: Deep Polarimetric Stereo Depth Estimation**. *International Conference on Computer Vision*. Published. 2023.
- **Zimo Wang**, Yicheng Wang, Sensen Wu, Zhenhong Du. **A House Price Valuation Model Based on Geographically Neural Network Weighted Regression**. *ISPRS International Journal of Geo-Information* 11, no. 8: 450. Published. 2022.

RESEARCH EXPERIENCES

- ByteDance – Generative AI Group** On-site
 - *Ph.D. Research Intern — Advisors: Gordon Guocheng Qian, Chongyang Ma* Summer 2026
 - **Video Diffusion Distillation**: Developed PDMD, a one-line projection that filters the student's training error from the distribution matching update, without auxiliary losses, additional networks, or extra model passes.
 - **Quantitative and User Evaluation**: Reached 83.73 VBench at 4 NFE on Wan2.1 (+1.03 vs. DMD), surpassing the 50-step teacher in VBench total score and user-rated visual and motion quality.
- Adobe Research – Computational Photography Group** On-site
 - *Ph.D. Research Intern — Advisors: Yuting Yang, Jiawen Chen* Summer 2025
 - **Tucker-Factorized LUT Prediction**: Proposed a Tucker-factorized formulation for predicting image-adaptive 3D LUTs, reducing parameterization while preserving flexible global color transformation capacity.
 - **Expressivity Analysis**: Investigated the expressivity limits of LUT-based enhancement and analyzed when complex color pipelines can be approximated within low-rank LUT subspaces.
 - **Low-Dimensional Structure in MIT Adobe FiveK**: Showed that retouch transformations in MIT Adobe FiveK exhibit an intrinsically low-dimensional structure under annotator variability, explaining the robustness of small-basis LUT mixtures.

- UCSD Center for Visual Computing** On-site
 • *Ph.D. Research Assistant — Advisor: Tzu-Mao Li* *Sep 2023 - Present*
 - Faster Image Generation Paradigm Producing Better Images:** Proposed *Distance Marching*, a distance-field-inspired, time-unconditional generation framework that predicts a distance-like scalar and denoising direction, enabling principled gradient-descent and sphere-tracing inference without explicit time inputs. Designed distance-focused objectives that reduce denoising-target ambiguity by emphasizing closer pairs, producing directions better aligned with the data manifold. Achieved consistent improvements: **~13.6%** lower FID on average than flow matching across backbone sizes, and exceeded flow matching's final FID using **~60%** of the sampling steps.
 - Innovative SDF Reconstruction via Physics-Inspired PDEs:** Discovered a novel loss for SDF reconstruction, fundamentally linked to a specific physical process and its corresponding PDEs. Solved the associated PDE to enhance SDF modeling, leading to groundbreaking performance beyond traditional eikonal equation-based methods. Achieved state-of-the-art results across multiple datasets.
 - Variational Optimization for Nondifferentiable Vector Graphics:** Developed an optimization framework for vector graphics based on variational optimization, enabling the differentiation of discrete decision-making problems such as shape type and layer order in vector graphics modeling. Proposed an approach to optimize discrete decisions in vector graphics, with ongoing research to integrate this framework with generative models for parameter space optimization and generation.
- IDEA Lab of University of Illinois at Urbana-Champaign** Remote
 • *Student Summer Research — Advisor: Hanghang Tong* *June 2022 - Oct 2022*
 - Time Series Long-term Prediction with Fourier Analysis:** Provided an innovative cross-correlation mechanism with the help of Fourier decomposition. Its correlation coefficients to several specific time lags can be rapidly solved with a complexity $N\log N$.
 - Dynamic Time Lags with Time Stamps:** Made the model learn the time lag relationships from time stamps, so that the geographical upstream and downstream relationships could be determined dynamically, and even some distant correlations could be examined.
- State Key Laboratory of CAD & CG of Zhejiang University** On-Site
 • *Student Research Assistant (Part-Time) — Advisor: Zhaopeng Cui* *Aug 2021 - Nov 2022*
 - SOTA Depth Estimation with Polarization Information:** Reconstructed depth with RGB and polarization stereo with a cascaded dual-GRU module and iso-depth constraints. The application of polarization images revealed a new task and enabled our model to outperform all the other depth estimation models.
 - Dataset Processing:** Participated in image collection, image synthesis, and dataset retrieval; explored the UE4 engine to render synthetic data and capture real images for our research.
 - Deep Involvement of Computer Vision Research:** Led presentations in the group meeting, grasped the whole process of computer vision research, independently authored the section of related works and modified the paper in line with amendments and suggestions.
- Zhejiang Provincial Laboratory of GIS** On-site
 • *Student Research Assistant (Part-Time) — Advisors: Sensen Wu, Zhenhong Du* *Mar 2021 - Apr 2022*